Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 31

The Multiplication and Addition Principles

Covers Chartrand & Zhang, Section 8.1

Status of this set

Ungraded practice. Solutions released with it. Show the product or sum, not just the number — the setup is where counting goes wrong.

Part I: How to Practice This Material

Which principle?

"And then" ⇒ multiply. **"Or," non-overlapping** ⇒ add.
Before multiplying, ask: *is the number of choices at each stage the same no matter what was chosen earlier?* If not, split into cases and add.

Three mistakes to expect

1. **Adding when you should multiply**, or the reverse.
2. **Forgetting a restriction** — a leading digit cannot be 0.
3. **Multiplying when a stage's count varies.** That calls for cases.

Part II: Practice Problems

Section A — Direct counts

1. Count each, showing the product or sum.

(a) Passwords of exactly 6 characters, each a lowercase letter or a digit.

(b) Three-digit numbers whose digits are all odd.

(c) Bit strings of length 10 that begin with 1 and end with 0.

2. How many four-digit numbers have no repeated digit? How many have *at least one* repeated digit?

Section B — Addition

3. How many bit strings have length at most 4? (Count length 1 through 4.)

Section C — When stages interfere

4. How many three-digit numbers are odd and have all distinct digits?

5. Explain why the count of three-digit *even* numbers with distinct digits cannot be computed as a single product, and then compute it.

Section D — Writing

6. In four to six complete sentences, explain how the Multiplication Principle relates to the Cartesian product from Lecture 11, and why the count of subsets of an n-element set is 2^n.